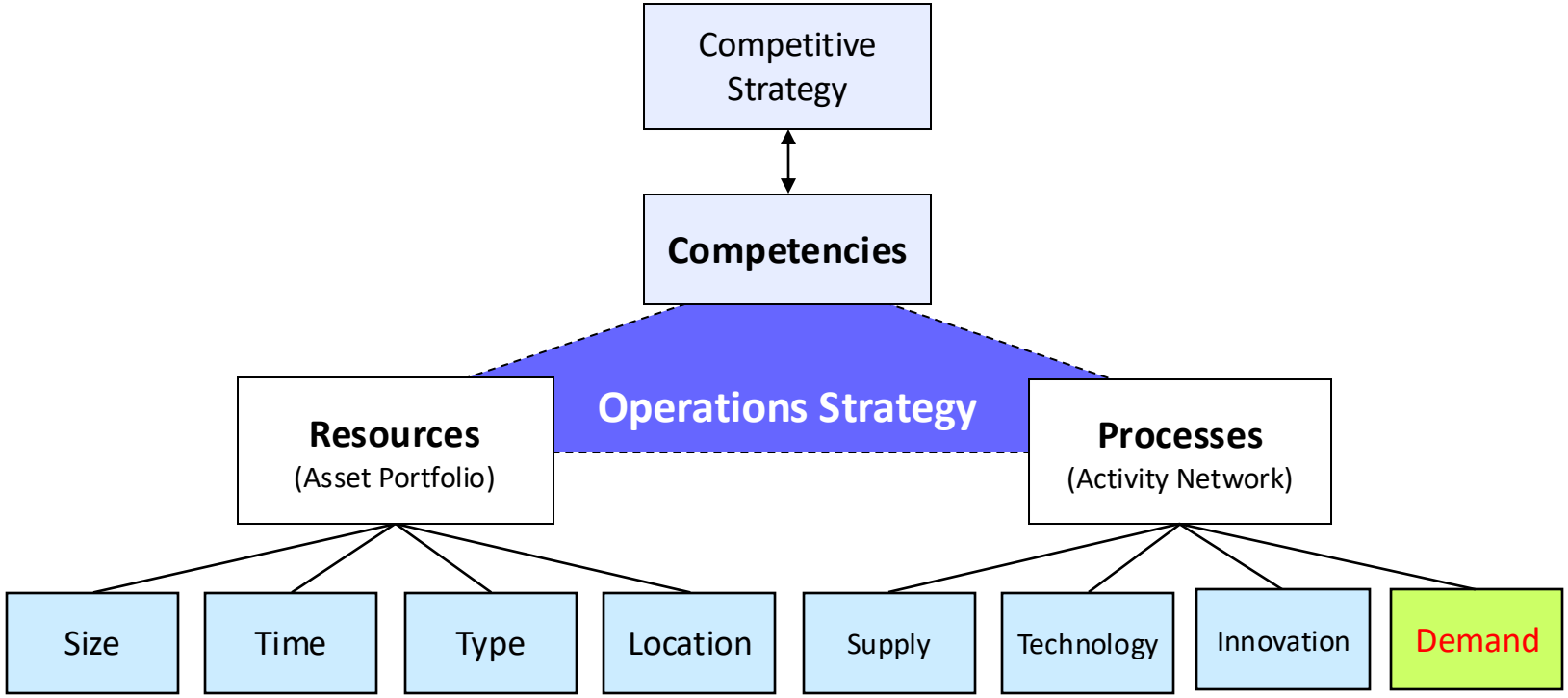
Operations Strategy Framework

In this class, we will discuss how to match supply with demand strategically. Specific topics include Revenue Management (RM).



Demand Management: Learning Objectives

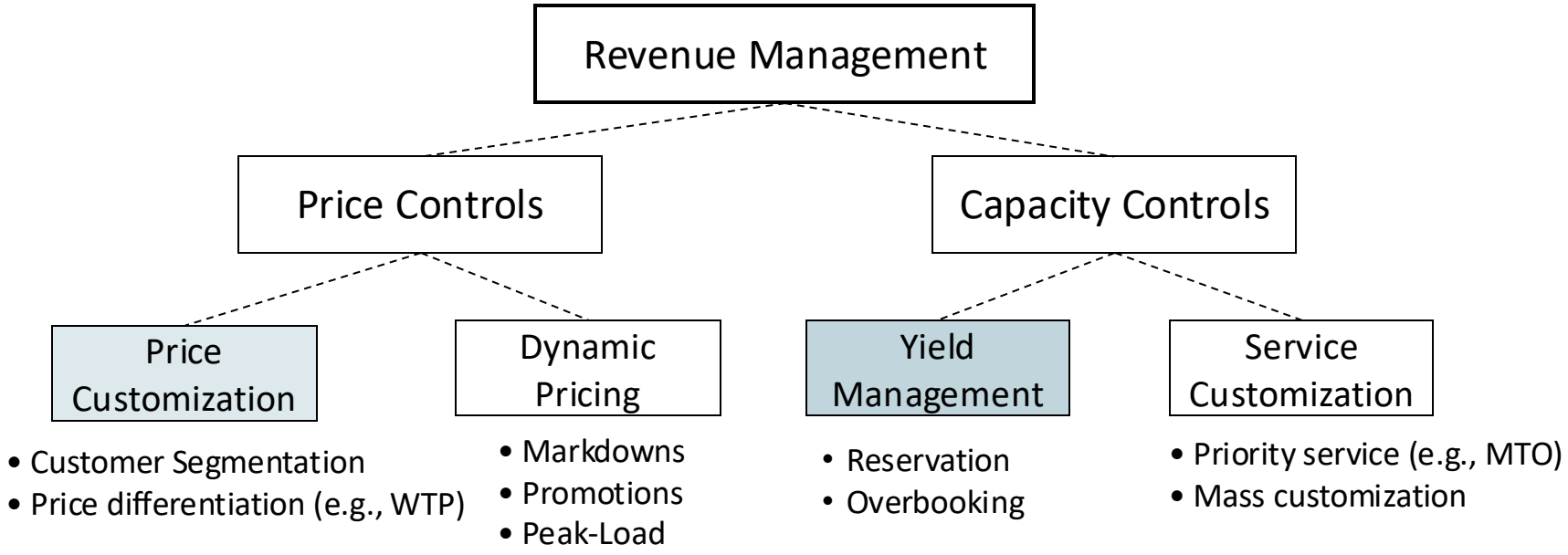
- Be able to
  - Explain the concept of revenue management (RM) and integrate it with operations strategy
  - Understand how to design products to achieve customer segmentation and price discrimination
    - Understand the operational counterpart of customer segmentation is capacity segmentation
      - Introduce *Littlewood's* rule for capacity allocation
  - Use marginal analysis for Yield management
  - Identify the conditions for and obstacles to effective revenue management

Demand Management

- Demand management includes all the activities that influence the demand for products and services, including marketing and sales activities.
- Operations can contribute to demand management, i.e., match supply and demand, by tailoring processes to demand characteristics, so to max **NPV**.
  - When existing operational system is inflexible, OS must match demand to available supply in a value-maximizing manner
- Revenue Management (RM) refers to the collection of demand management practices that aim to maximize the revenue of available supply, it should be integrated and aligned with overall operation strategy (OS) -
  - Evaluate company competitive strategy and customer: what do customers need and value? Do valuations change significantly over customers over time?
    1. If yes, there is an opportunity to serve them better and improve profits by designing products tailored to each customer group and the time period
    2. Forecast demand, price products based on customer valuations and behavior
    3. Control capacity that is allocated to the various products

Revenue Management

- RM practices can accordingly be categorized as processes that aim to improve revenue by predominantly controlling price or capacity or both
  - Price controls are techniques that vary prices to influence demand
  - Capacity controls are techniques that restricts the available supply to influence demand



Price customization

- Ideally, we would like to perfectly discriminate (sort) customers based on their WTP or reservation price, and charge them exactly that amount (1st degree price discrimination); the problem is that it is usually unobservable
  - Solution: design *differentiated* "products" according to a set of features and fences and let customers choose, thereby partially revealing their WTP (2nd degree price discrimination)
    - The trick is to match product features to customer preferences and WTP (customer segmentation)
    - Finding a good differentiation method is often an art than a science
  - Examples:
    - Time of purchase or usage or both: advance, spot, day-of-week, time of the day, etc.
    - Purchase restrictions: cancellation options, Saturday-night-stay, etc.
    - Duration of usage: single-day-rate, weekly-rate, monthly-rate
    - Purchase volume: individual vs. group
    - Customer affiliation: corporate rate
    - A "product" is also called a "fare-class" (e.g., Class Q, E, G in the airline industry)

Price customization

- Effective product designs uses features and fences that strongly correlate with customer preferences and reservation prices
- When should we consider price customization, or customer segmentation and differentiating pricing?
  1. The valuation of an identical asset unit varies considerably over customers
  2. Customers can be efficiently and effectively segmented into groups with similar WTP
  3. Customer valuation of each segment can be forecasted with reasonable accuracy

Capacity Segmentation and Allocation

- The operational counterpart of customer segmentation is capacity segmentation
  - Segment the same capacity (e.g., economy cabin) into different buckets
  - Allocating capacity (of the different buckets) to different "products" and demands (for these different "products" that consume the same capacity) that come in over time
- How? Examples of capacity allocation mechanisms:
  - Ad-hoc negotiation
  - Posting remaining availability (reservation system booking limits)
  - Open/closed status indicators
  - Bid prices ("minimum acceptable fare")
- Ultimately, this boils down to an *accept/deny* decision for each demand requests
  - Rule: accept request if Revenue > Displacement Cost (*opportunity cost*)

Capacity Segmentation and Allocation

<u>Example</u>

- Consider the 5pm-flight from Singapore to Brussels 6 months from now
- Economy seat capacity K = 300 seats
- Two "fare classes" with posted prices: p_1= \$2,400 and p_2= \$600
- Demand for each "fare class":
  - D_1 (for p_1) is *Normal* with μ= 50 and σ= 20
  - D_2 *(for p_2)* has ample demand (exceeds capacity of 300 seats)
- Demand for low fare is realized *before* high fare demand (advance booking)
- How many seats, Q, should we protect from advance sales at \$600?
- We deny advance booking at \$600 if we can sell it later at \$2400, means if 2400*Prob(D_1 > Q) > 600
- We should protect: Q = norminv($\frac{2400-600}{2400}$, 50, 20) = 63

Capacity Segmentation and Allocation

- An example of 3 fare classes with total capacity of 300 seats:

Class	Price	Demand Mean	SD
A	2400	25	5.3
B	800	50	8.5
C	600	Ample demand	

- We can estimate the optimal protection levels for class A and B against class C by using Littlewood's rule
  - At start of stage C, the optimal protection level for class A against class C:
    - We deny advance booking (to class C) if \$600 < \$2400 × Prob(we sell at class A), means if Prob(we sell at class A) > 600/2400=25%, we deny. Optimal seats should be "protected" for class A = norminv(75%, 25, 5.3) ≈ 29
  - Also, the optimal protection level for class B against class C:
    - we deny advance booking (to class C) if \$600 < \$800 × Prob(we sell at class B), means if Prob(we sell at class B) > 600/800=75%, we deny. So, Seats should be "protected" for class B = norminv(25%, 50, 8.5) ≈ 44
  - So, total protection level for classes A and B against C ≈ 73 (=29+44)
    - Alternatively, the "booking limit" for class C is 300 – 73 ≈ 227
  - At the start of stage B, assume you have 80 seats left, then you may recompute the protection level for class A against class B:
    - We deny advance booking (to class B) if \$800 < \$2400 × Prob(we sell at class A), means if Prob(we sell at class A fare) > 800/2400 = 33.33%, we deny. Seats should be "protected" for class A = norminv(66.67%, 25, 5.3) ≈ 27. Alternatively, the "booking limit" for class B ≈ 53

Example of Capacity Reservation

Decision: to maximize its expected profits, how much floating capacity should Solectron reserve at its Penang plant each month?

- Solectron, a leading provider of manufacturing services to electronics OEM's worldwide, can produce up to 27,000 circuit boards per month in its Penang, Malaysia fabrication (fab) plant.
- It has two options for generating revenue from this fab capacity:
  1. It can accept a proposed contracts (e.g., from Intel, before demand is known) that would generate profits of $60 per circuit board, or
  2. It could reserve some of its capacity as "floating". This un-contracted capacity could worth $100 per circuit board, if Intel (or other customers) need that capacity in short term without pre-contract, i.e., after demand is known.
- Past few years' data have indicated that monthly demand for "floating" circuit boards from short-term customers (those without pre-contracts) has a normal distribution with a mean of 7,000 and a standard deviation of 1,000, i.e., Demand for floating capacity is random and known as Normal ~ $N(7000, 1000^2)$.

Example of Overbooking

- Registration for art classes at an art studio allows overbooking. The studio's typical classrooms has 50 seats each profits $100. If overbooks, it costs $200 to compensate the customer whose reservation cannot be honored. Based on the past data, the estimated no-shows have a discrete distribution as shown in the following table. What's the optimal number of overbookings?

No-shows (x)	Probability p	Cumulative Prob(X≤x)=F(x)
1	0.30	0.30
2	0.34	0.64
3	0.08	0.72
4	0.07	0.79
5	0.06	0.85
6	0.05	0.9
7	0.04	0.94
8	0.03	0.97
9	0.02	0.99
10	0.01	1.0

When consider Revenue Management?

- Constrained, perishable supply
  - Perishable “product”
  - Constrained supply (available product is limited and demand may exceed supply)
  - Sales season is limited
- Demand is dynamic and volatile
  - Pricing and product allocation decision must be made before uncertainty in demand is resolved.
- Customer Segmentation is possible
  - Valuation varies considerably over the customer base so that there is an opportunity to segment customers (so that different prices can be charged)
  - We have sorting variables to segment customers
  - We can forecast valuation of each segment reasonably well
- Example:
  - The same unit of capacity (e.g., the same economy airline seat) can be used to deliver services to different customer segments (e.g., business and leisure customers) at different prices (and the marginal unit cost is roughly unaffected)

Revenue Management Obstacles

- Data mining, forecasting, and optimization
- Competition and automated revenue management
- Treat your (best) customers well
- Social concerns and fairness (non-market factors)
- …